Transmission media

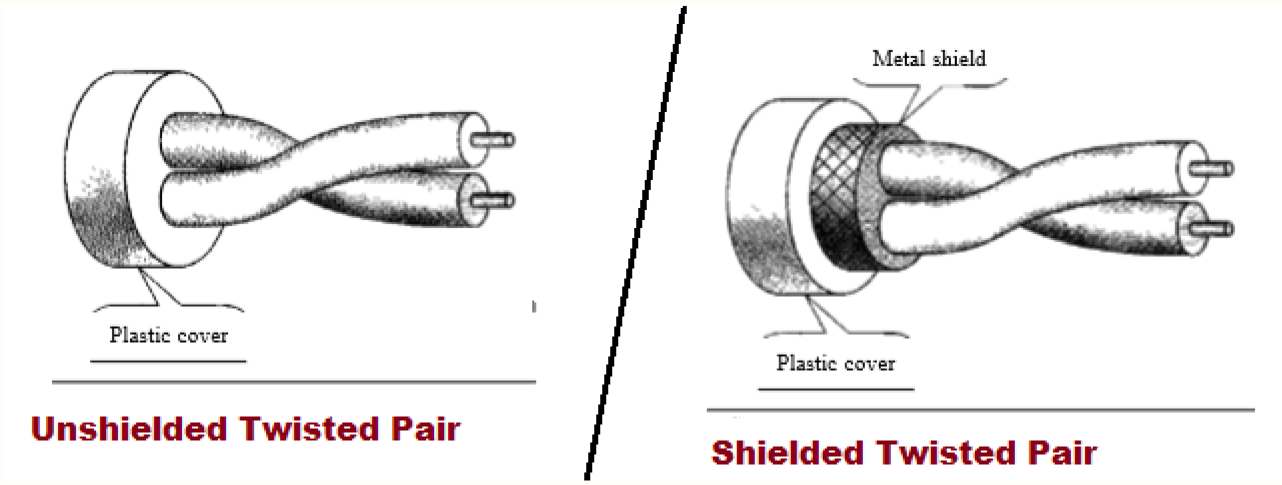
Transmission media *is the physical path through which data signals travel from sender to receiver. It is broadly divided into* ***guided media*** *(where the signal is confined within a physical conductor, like cables) and* ***unguided media*** *(where the signal travels freely through air or space, like radio waves). The choice of medium affects the speed, distance, and quality of communication.*

1. Guided transmission media

Guided transmission media are those in which the signal is physically confined and directed through a solid medium. There are three main types.

I. Twisted Pair Cable

A ***twisted pair cable*** *consists of two insulated copper wires twisted around each other. The twisting helps cancel out electromagnetic interference (EMI) from external sources and from neighboring pairs — this effect is called* ***noise cancellation through cancellation of induced EMF****.*



UTP (Unshielded Twisted Pair)

UTP has no additional shielding beyond the basic plastic insulation around each wire. It is the most commonly used cable in LANs and telephone networks.

Advantages:

- Cheap and easy to install.
- Lightweight and flexible, making it ideal for indoor wiring.
- Sufficient for most LAN applications (up to Gigabit Ethernet with Cat6).

Disadvantages:

- More susceptible to EMI and crosstalk compared to STP.
- Limited bandwidth and shorter effective range.
- Not suitable for environments with heavy electrical interference.

STP (Shielded Twisted Pair)

STP has an additional metallic foil or braided shield around each pair or around the entire cable bundle, providing extra protection against interference.

Advantages:

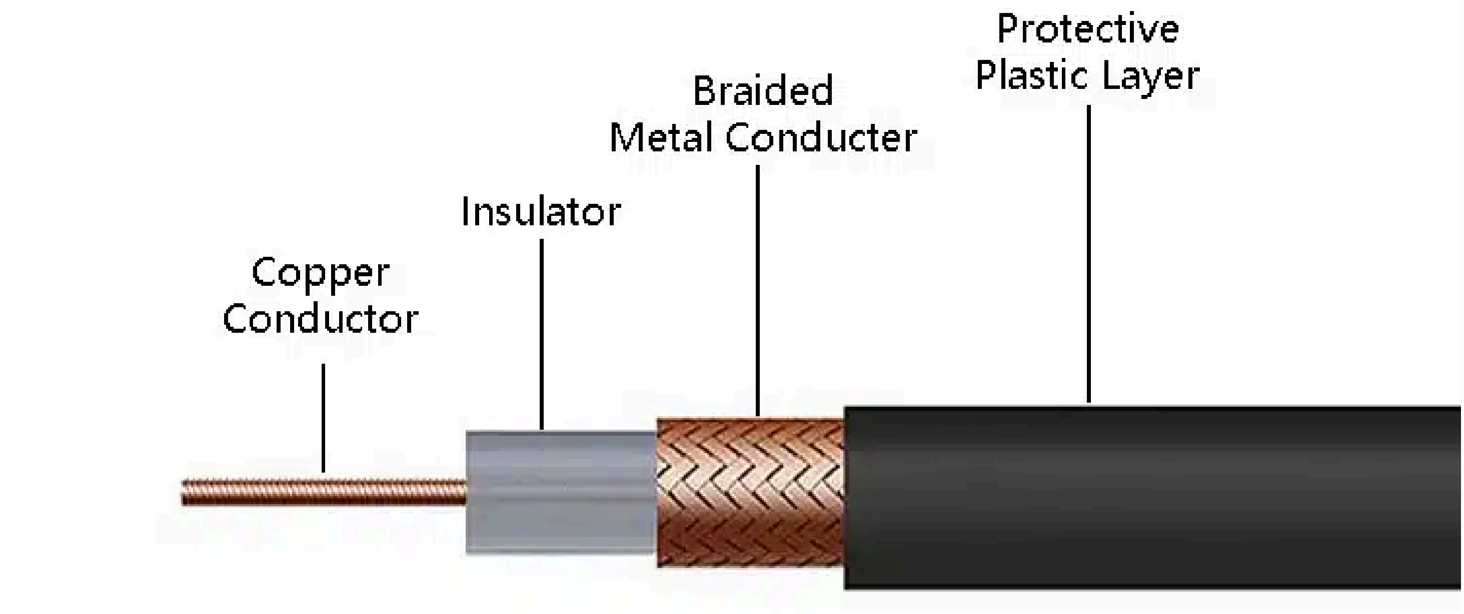
- Much better noise immunity than UTP due to the shielding.
- Supports higher data rates over longer distances.
- Suitable for electrically noisy environments like factory floors.

Disadvantages:

- More expensive and heavier than UTP.
- Harder to install — the shield must be properly grounded, otherwise it can act as an antenna and worsen interference.
- Bulkier, making it less flexible for tight cable runs.

2. Coaxial Cable

A **coaxial cable** carries signals through a central copper conductor surrounded by concentric layers — an insulating material, a metallic shield (braided or foil), and an outer plastic jacket. The term "coaxial" refers to the fact that all layers share the same axis. This layered structure gives it far better noise immunity and higher bandwidth than twisted pair.



Structure (inside out)

- **Inner conductor** — carries the signal (solid or stranded copper).
- **Dielectric insulator** — separates the inner conductor from the shield.
- **Metallic shield** — blocks external EMI and prevents signal leakage.
- **Outer plastic jacket** — provides physical protection.

Advantages

- Much higher bandwidth and noise immunity than twisted pair.
- The shielding makes it resistant to EMI and crosstalk.
- Suitable for both analog (cable TV) and digital (broadband internet) signals.
- Can carry signals over longer distances without significant attenuation.

Disadvantages

- More expensive and bulkier than twisted pair.
- Harder to install and less flexible, especially thick coax.
- Largely replaced by fiber optics for high-speed long-distance links.

Applications

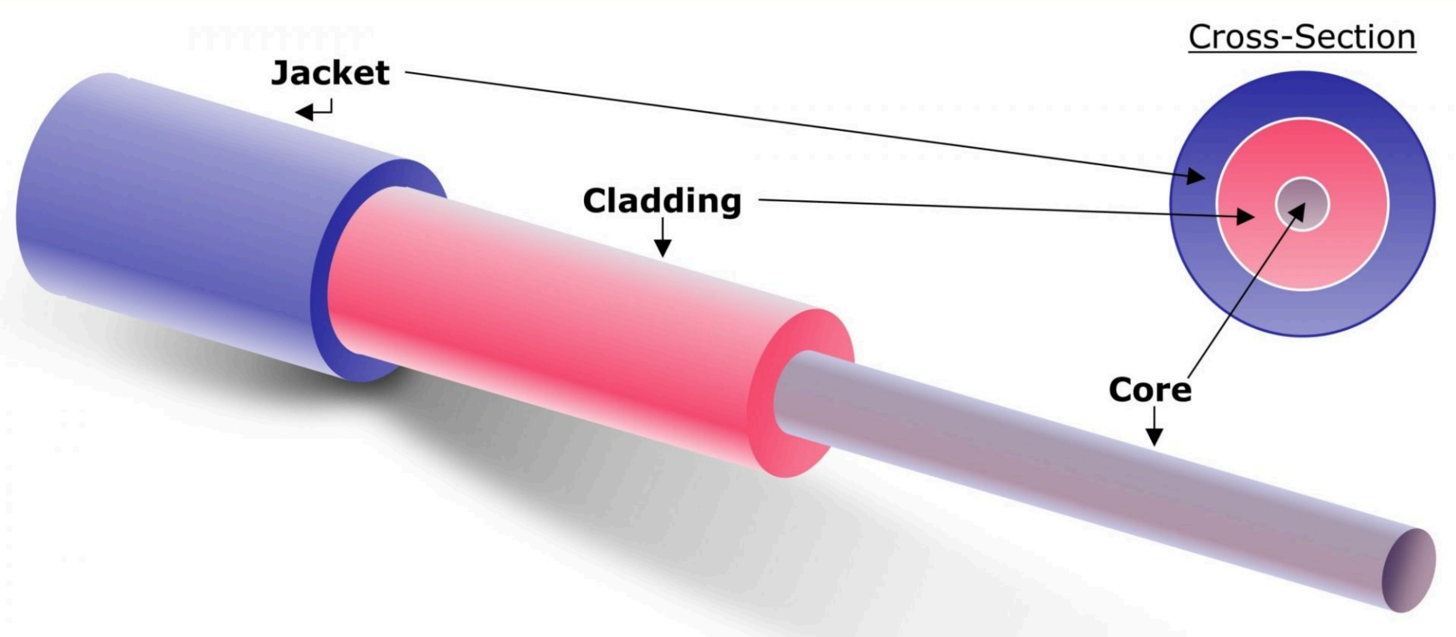
- *Cable television (CATV) distribution networks.*
- *Broadband internet (cable internet).*
- *Previously used in Ethernet LANs (now mostly replaced by UTP and fiber)*

3. Fiber Optic Cable

*A **fiber optic cable** transmits data as pulses of light through a thin glass or plastic core using the principle of **total internal reflection** — light entering the core at a certain angle bounces along the fiber without escaping. It offers the highest bandwidth, lowest attenuation, and complete immunity to electromagnetic interference among all guided media.*

Structure (inside out)

- ***Core** — thin glass or plastic through which light travels.*
- ***Cladding** — surrounds the core with a lower refractive index, causing total internal reflection.*
- ***Jacket** — provides physical protection.*



Modes of Transmission

Single-Mode Fiber (SMF)

In **single-mode fiber**, the core diameter is very small (around 8–10 μm), allowing only one mode (path) of light to travel straight through the core with no reflection. This eliminates **modal dispersion**, allowing signals to travel over very long distances at very high speeds. It uses a laser as the light source and is used in long-distance telecommunications and internet backbone links.

Multimode Fiber (MMF)

In **multimode fiber**, the core diameter is much larger (50–62.5 μm), allowing multiple modes (paths) of light to travel simultaneously at different angles. This causes **modal dispersion** — different light paths arrive at slightly different times, limiting distance and bandwidth. It uses an LED as the light source and is used in short-distance applications like LAN connections within a building.

Multimode further divides into two types:

Step-Index Multimode	Graded-Index Multimode
The refractive index changes abruptly at the core-cladding boundary.	The refractive index decreases gradually from center to edge of the core.
High modal dispersion — rays traveling longer paths slow down the signal significantly.	Lower modal dispersion — outer rays travel faster and arrive roughly at the same time as central rays.
Poor performance over long distances.	Better performance than step-index, suitable for moderate distances.

Advantages

- Extremely high bandwidth compared to copper media.
- Very low signal attenuation over long distances.
- Completely immune to EMI and crosstalk.
- More secure — signals do not radiate outside the cable.

Disadvantages

- More expensive to install than copper cables.
- Fragile — glass core can break under sharp bends or physical stress.
- Requires specialized equipment and skill for splicing and termination.
- Only carries optical signals, so electrical-to-optical conversion is needed.

	Twisted Pair	**Coaxial Cable**	**Fiber Optic**
Signal type	Electrical	Electrical	Light (optical)
Bandwidth	Low to moderate	Moderate to high	Very high
Attenuation	High	Moderate	Very low
EMI immunity	Low (UTP) / Moderate (STP)	High	Complete immunity
Distance	Short (up to ~100m for LAN)	Moderate (up to 500m)	Very long (km range)
Cost	Cheapest	Moderate	Most expensive
Installation	Easy and flexible	Moderate, less flexible	Difficult, requires special tools
Security	Low — signal leaks easily	Moderate	High — no signal radiation
Typical use	LAN, telephone networks	Cable TV, broadband internet	Backbone links, long-distance telecom
Advantages	Cheap, lightweight, easy to install, widely available	Better noise immunity than twisted pair, supports long runs, handles both analog and digital	Highest bandwidth, lowest attenuation, immune to EMI, very secure

	Twisted Pair	Coaxial Cable	Fiber Optic
Disadvantages	Susceptible to EMI and crosstalk, limited range and bandwidth	Bulky, expensive compared to twisted pair, mostly replaced by fiber	Fragile, costly to install, requires electrical-to-optical conversion

2. Unguided Transmission Media

In **unguided media**, signals travel freely through air, water, or vacuum without any physical conductor. The signal is broadcast as electromagnetic waves, and the transmission mode depends on the frequency used.

Propagation Methods / Types of Radio Wave Propagation

1. Ground Wave Propagation

In **ground wave propagation**, radio waves travel along the surface of the earth, following its curvature. It is effective at low frequencies (below 3 MHz) and can cover distances up to a few thousand kilometers. AM radio and maritime communication use this method.

2. Sky Wave Propagation

In **sky wave propagation**, signals are transmitted upward at an angle and reflected back to earth by the **ionosphere**. This allows communication over very long distances without relay stations. It works best in the frequency range of 3–30 MHz and is used in shortwave radio and international broadcasting. The ionosphere acts as a natural mirror for these frequencies.

3. Line-of-Sight (LOS) Propagation

In **LOS propagation**, signals travel in a straight line directly between transmitter and receiver. Since the signal does not bend or reflect, both antennas must be within visual range of each other. It is used at very high frequencies (above 30 MHz) including microwave and satellite links. Tall towers and relay stations are used to extend range by compensating for earth's curvature.

4. Space Wave Propagation

__Space wave propagation__ is a form of LOS where signals travel through the troposphere — the lowest layer of the atmosphere. It includes both direct waves and ground-reflected waves. It is used in VHF/UHF communication, FM radio, and television broadcasting.

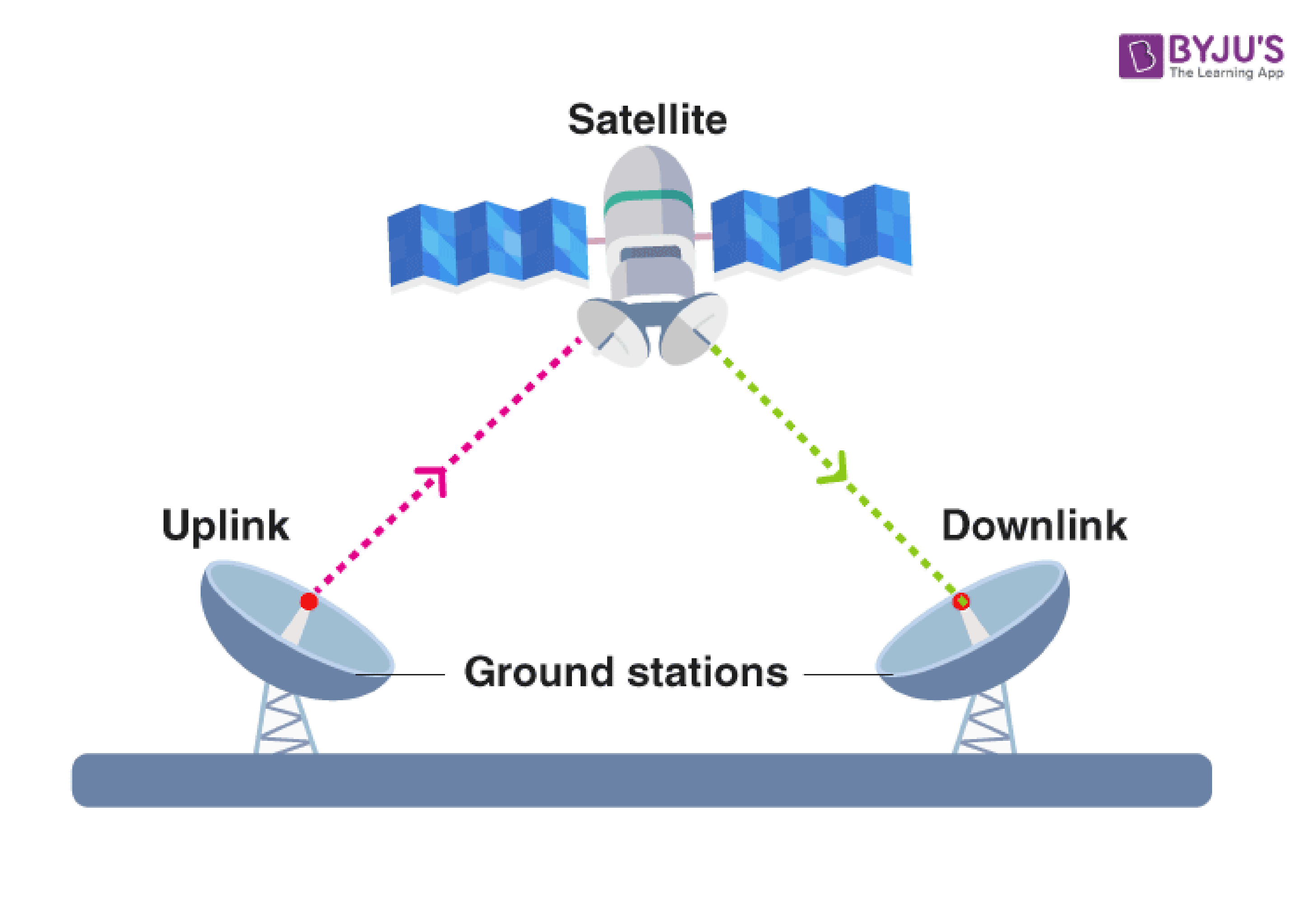
Types of Unguided Media

1. Terrestrial Microwave

__Terrestrial microwave__ uses high-frequency radio waves (1–40 GHz) transmitted between dish antennas mounted on towers or tall buildings. Since it is LOS, relay stations are placed every 20–80 km to extend range. It is used for telephone backbones and point-to-point data links.

2. Satellite Communication

A __satellite communication system__ uses an artificial satellite orbiting the Earth as a relay station to transmit signals between ground stations. It enables long-distance communication over areas where laying cables is impractical — oceans, mountains, remote regions.



Components

1. Space Segment (Satellite)

The satellite itself, orbiting in space. Contains a ***transponder*** *that receives the uplink signal, amplifies it, shifts its frequency, and retransmits it as a downlink signal.*

2. Ground Segment (Earth Station)

The ground-based equipment that communicates with the satellite. Consists of a ***dish antenna****, transmitter, receiver, and signal processing equipment. There are two types —* ***uplink station*** *(transmits to satellite) and* ***downlink station*** *(receives from satellite).*

3. Control Station

Monitors and controls the satellite's position, orientation, and health. Sends ***telemetry, tracking, and command (TT&C)*** *signals to keep the satellite on its correct orbital path.*

3. Cellular System

A ***cellular system*** *divides a geographic area into small regions called* ***cells****, each served by a base station. As a user moves between cells, the connection is handed off from one base* station to the next — this is called a **handoff**. It is the foundation of mobile phone networks (2G, 3G, 4G, 5G).

	Guided	Unguided
Signal path	Confined within a physical medium	Travels freely through air or space
Medium	Twisted pair, coaxial, fiber optic	Air, vacuum, water
Installation	Requires physical cabling	No physical cables needed
Cost	Higher installation cost	Lower infrastructure cost
Mobility	Fixed — devices must be wired	Supports mobile and remote devices
Security	More secure — signal stays in cable	Less secure — signal can be intercepted
EMI	Susceptible (except fiber)	Highly susceptible to interference
Bandwidth	High (especially fiber)	Moderate to high depending on frequency
Distance	Limited by cable length and attenuation	Can cover very large distances (e.g. satellite)
Example	LAN cables, fiber backbone	Wi-Fi, satellite, cellular networks

Data Rate Limits

The **data rate** of a channel defines how fast data can be transmitted. Two fundamental theorems set the theoretical maximum — **Nyquist's theorem** for ideal noiseless channels and **Shannon's theorem** for real noisy channels.

Nyquist Bit Rate (Noiseless Channel)

Nyquist's theorem *states that for a noiseless channel, the maximum data rate is determined by the bandwidth and the number of discrete signal levels used.*

Formula

$$\text{Bit Rate} = 2 \times B \times \log_2 M$$

Where:

- ***B*** *= bandwidth of the channel (in Hz)*
- ***M*** *= number of discrete signal levels (voltage levels)*
- ***$\log_2$ M*** *= number of bits per signal level*

Explanation

A signal with bandwidth B can change at most ***2B times per second*** *— this is the Nyquist rate. Each change (sample) carries $\log_2$ M bits depending on how many levels the signal uses. Multiplying both gives the maximum achievable bit rate.*

Limit Cases

- *If* ***M = 2*** *(binary), bit rate = 2B — the simplest case with only two levels (0 and 1).*
- *Increasing M increases the bit rate, but in practice, more levels make the signal harder to distinguish, especially with noise.*
- *Nyquist assumes a* ***perfect, noiseless*** *channel — purely theoretical.*

Example

A noiseless channel with bandwidth 3000 Hz using 8 signal levels:

$$\text{Bit Rate} = 2 \times 3000 \times \log_2 8 = 2 \times 3000 \times 3 = 18{,}000 \text{ bps}$$

Shannon Capacity (Noisy Channel)

Shannon's theorem *gives the maximum data rate of a channel in the presence of noise, defined by the* ***Signal-to-Noise Ratio (SNR)****.*

Formula

$$C = B \times \log_2(1 + \text{SNR})$$

Where:

- **C** = channel capacity (maximum bit rate in bps)
- **B** = bandwidth of the channel (in Hz)
- **SNR** = Signal-to-Noise Ratio (ratio of signal power to noise power, dimensionless)

SNR is often expressed in decibels:

$$SNR_{dB} = 10 \times \log_{10}(SNR)$$

Explanation

No matter how many signal levels are used, noise limits how reliably the receiver can distinguish them. Shannon's formula captures this fundamental limit — beyond capacity C, error-free communication is impossible regardless of the encoding scheme used.

Limit Cases

- If **SNR = 0** (noise equals signal), then $C = B \times \log_2(1) =$ **0** — no information can be transmitted.
- If **SNR** → ∞ (no noise), capacity is theoretically infinite — which aligns with Nyquist (a noiseless channel is only limited by bandwidth and levels).
- Higher bandwidth B increases capacity, but doubling B does not double capacity if SNR is low.

Example

A telephone channel with bandwidth 3000 Hz and SNR = 1023:

$$C = 3000 \times \log_2(1 + 1023) = 3000 \times \log_2(1024) = 3000 \times 10 = 30{,}000 \text{ bps}$$

Nyquist vs Shannon

	Nyquist	**Shannon**
Channel assumed	Noiseless	Noisy

	Nyquist	Shannon
Formula	$2 \times B \times \log_2 M$	$B \times \log_2(1 + SNR)$
Controlled by	Bandwidth and signal levels	Bandwidth and SNR
Practical?	Theoretical upper bound	More realistic upper bound
Key insight	More levels = more bits per sample	More noise = less capacity

In practice, both theorems are used together — Nyquist tells you what is possible with ideal signal levels, and Shannon tells you what noise allows.